

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                                          |
|------------|------|---------------|-------------------------------------------------------------------------------|-----------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Medium <div><div></div><div></div><div></div></div> |

Question

$$52(x^3 + 64)(x^4 - 81) = 0$$

How many distinct real solutions does the given equation have?

Answer

- A. Exactly two
- B. Exactly three
- C. Exactly five
- D. Exactly seven

Correct Answer: B

Rationale

Choice B is correct. The given equation can be rewritten as  $52(x + 4)(x^2 - 4x + 16)(x^2 - 9)(x^2 + 9) = 0$ , which is equivalent to  $52(x + 4)(x^2 - 4x + 16)(x - 3)(x + 3)(x^2 + 9) = 0$ . Applying the zero product property in this equation yields  $52 = 0$ ,  $x + 4 = 0$ ,  $x^2 - 4x + 16 = 0$ ,  $x - 3 = 0$ ,  $x + 3 = 0$ , and  $x^2 + 9 = 0$ . The equation  $52 = 0$  has no solution. Subtracting 4 from both sides of the equation  $x + 4 = 0$  yields  $x = -4$ . Adding 3 to both sides of the equation  $x - 3 = 0$  yields  $x = 3$ . Subtracting 3 from both sides of the equation  $x + 3 = 0$  yields  $x = -3$ . The discriminant of a quadratic equation of the form  $ax^2 + bx + c = 0$  is  $b^2 - 4ac$ . If the discriminant is positive, the equation has two distinct real solutions. If the discriminant is zero, the equation has one distinct real solution. If the discriminant is negative, the equation has no real solutions. In the equation  $x^2 - 4x + 16 = 0$ ,  $a = 1$ ,  $b = -4$ , and  $c = 16$ . Substituting these values in the equation for the discriminant,  $b^2 - 4ac$ , yields  $(-4)^2 - 4(1)(16)$ , or  $-48$ . Since the discriminant is negative, this equation has no real solutions. In the equation  $x^2 + 9 = 0$ ,  $a = 1$ ,  $b = 0$ , and  $c = 9$ . Substituting these values in the equation for the discriminant,  $b^2 - 4ac$ , yields  $0^2 - 4(1)(9)$ , or  $-36$ . Since the discriminant is negative, this equation has no real solutions. Therefore, the distinct real solutions to the given equation are  $x = -4$ ,  $x = 3$ , and  $x = -3$ . Thus, the given equation has exactly three distinct real solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.